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Molded conduit penetration seal

Abstract

A conduit seal that serves to connect conduits between different environments while providing sufficient insulation to mitigate condensation and molding issues. The conduit seal can also provide grounding continuity with an electrical conducting insert, thereby providing electrical continuity across a thermal barrier between the environments. The conduit seal includes a through-hole connector including a body, a first end, and a second end, a compliant member, and a locking member. The body has an exterior surface, and the body defines an internal cavity. The compliant member is disposed circumferentially around the exterior surface of the through-hole connector. The compliant member is connected to the locking member and is configured to seal the locking member to a barrier.

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Background/Summary

FIELD

(1) This disclosure relates generally to a conduit seal extending between two environments. More particularly, this disclosure relates to a conduit seal extending between two environments, which reduces thermal conductivity while optionally maintaining electrical continuity.

BACKGROUND

(2) In certain industries, a conduit assembly is desired to extend between two different places.

SUMMARY

(3) In some embodiments, a conduit seal comprises a through-hole connector, wherein the through-

hole connector includes a body, wherein the body has an exterior surface, wherein the body defines an internal cavity, a first end, and a second end, wherein the second end is longitudinally opposite the first end; a compliant member, wherein the compliant member is disposed circumferentially around the exterior surface of the through-hole connector; and a locking member, wherein the compliant member is connected to the locking member and is configured to seal the locking member to a barrier.

(4) In some embodiments, the conduit seal further comprises a second compliant member, wherein the second compliant member is disposed circumferentially around the exterior surface of the through-hole connector.

(5) In some embodiments of the conduit seal, the second end of the through-hole connector comprises a flange, and the second compliant member is connected to the flange and is configured to seal the flange to the barrier.

(6) In some embodiments, the conduit seal further comprises at least one electrical conducting insert, wherein the at least one electrical conducting insert extends longitudinally within the internal cavity along the through-hole connector to electrically connect the first end of the through-hole connector to the second end of the through-hole connector.

(7) In some embodiments of the conduit seal, the at least one electrical conducting insert comprises a first interface member connected to the first end of the through-hole connector; a second interface member connected to the second end of the through-hole connector; and a conducting member electrically connecting the first interface member to the second interface member.

(8) In some embodiments of the conduit seal, the first interface member has a first threaded portion and the second interface member has a second threaded portion, wherein the first threaded portion of the first interface member is configured to be connected to a first, separate conduit, and wherein the second threaded portion of the second interface member is configured to be connected to a second, separate conduit.

(9) In some embodiments of the conduit seal, the first interface member and the second interface member comprise a metallic material.

(10) In some embodiments of the conduit seal, the conducting member comprises copper, aluminum, or stainless steel.

(11) In some embodiments of the conduit seal, the compliant member and the second compliant member are elastomeric.

(12) In some embodiments of the conduit seal, the compliant member and the second compliant member comprise polyisoprene, polybutadiene (butadiene rubber), styrene-butadiene rubber, acrylonitrile-butadiene copolymer, isobutylene-isoprene copolymer, ethylene-propylene copolymer, polychloroprene, polysulfide, silicones (e.g., polydimethylsiloxane), polyurethane, polyacrylate elastomer, chlorinated polyethylene, styrene-isoprene-styrene copolymer (SIS), styrene-butadiene-styrene copolymer (SBS), ethylene-propylene-diene polymer, fluoroelastomers (e.g., tetrafluoroethylene propylene, fluorosilicone, and perfluoroelastomer), or any combination thereof.

(13) In some embodiments of the conduit seal, the through-hole connector is injection molded, compression molded, transfer molded, or potted.

(14) In some embodiments of the conduit seal, the through-hole connector comprises a thermoplastic or a thermoset resin.

(15) In some embodiments of the conduit seal, the through-hole connector comprises an additive of ultra-violet (UV) resistance, antimicrobial, conductivity, mechanical and thermal stability, chemical resistance, insulation property or decreased flammability.

(16) In some embodiments of the conduit seal, a thermal conductivity of the through-hole connector ranges from about 0.2 Watt per meter-Kelvin to about 5 Watt per meter-Kelvin.

(17) In some embodiments, a method of securing a conduit seal to a barrier comprises inserting a first end of a through-hole connector through the barrier to a first side of the barrier; maintaining a flange connected to a second end of the through-hole connector on a second side of the barrier;

sealing the through-hole connector to the barrier by connecting a first compliant member connected to a locking member to the first side of the barrier and connecting a second compliant member connected to the flange to the second side of the barrier.

(18) In some embodiments of the method of securing the conduit seal to the barrier, the barrier is a floor in a building, and wherein the first side of the barrier is located vertically above the second side of the barrier.

(19) In some embodiments of the method of securing the conduit seal to the barrier, the barrier is a floor in a building, and wherein the second side of the barrier is located vertically above the first side of the barrier.

(20) In some embodiments of the method of securing the conduit seal to the barrier, the barrier is a wall in a building.

(21) In some embodiments of the method of securing the conduit seal to the barrier, the locking member has a threaded interior surface and the first end of the through-hole connector has a threaded exterior surface, wherein the threaded interior surface of the locking member and the threaded exterior surface of the first end of the through-hole connector are reciprocally threaded, and wherein the connecting the first compliant member to the first side of the barrier comprises screwing the threaded interior surface of the locking member onto the threaded exterior surface of the first end of the through-hole connector.

(22) In some embodiments, the method of securing the conduit seal to the barrier further comprises connecting a first conduit to a first end of the conduit seal.

(23) In some embodiments, the method of securing the conduit seal to the barrier further comprises connecting a second conduit to a second end of the conduit seal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) References are made to the accompanying drawings that form a part of this disclosure and that illustrate embodiments in which the systems and methods described in this Specification can be practiced.

(2) FIG. 1A-1F show an exemplary conduit seal according to some of the embodiments.

(3) FIG. 2 shows an exemplary conduit seal according to some of the embodiments.

(4) FIG. 3 shows a conduit seal according to some of the embodiments.

(5) FIG. 4 shows an exemplary flowchart according to some of the embodiments for securing a conduit seal to a barrier.

(6) Like reference numbers represent the same or similar parts throughout.

DETAILED DESCRIPTION

(7) When conduits extend continuously across a change in environment, there can be a heat leak or gain. The change in environment can increase heating, venting, and air conditioning (HVAC) requirements and also present an issue of unwanted condensation. One example is where the conduit penetrates through an insulation wall from a cold environment to the relatively warm side. Without wishing to be bound by theory, the conduit surface is cold due to conductive heat transfer from the colder side of the conduit. The cool surface then condenses water in the warmer environment. The wet surface then provides an opportunity for mold or bacteria to grow. To block moisture penetration and mitigate cooling effect, users will often fill the void between the conduit and the through hole with flexible polymer caulking. This seal, however, will often loosen with time, increasing convection or passage of other contaminants (bugs).

(8) One of the currently employed solutions involves a continuous conduit and use of an elastomer to fill the gap between conduit and structure. In some examples, a polyvinyl chloride (PVC) nipple has been employed to provide a thermal barrier between sides. However, the PVC nipple does not

provide grounding continuity (between commonly used metallic conduit systems) and does not provide a reliable sealing along the conduit outer diameter (OD) and structure.

(9) The present disclosure describes a conduit seal that serves as the connection of conduits between different environments (such as different temperature and relative humidity) while providing sufficient insulation to mitigate condensation and molding issues. This hygienic design helps eliminate germination and contamination. The conduit seal can also provide grounding continuity with an electrical conducting insert, thereby providing electrical continuity across the thermal barrier between the environments. The present disclosure can be used in a variety of applications including the food and beverage industry (F&B) and the pharmaceutical industry. More generally, the present disclosure may be used where a conduit passes between environments of different temperatures or other conditions. For example, one exemplary application is when a conduit passes from an interior space to an exterior space. Segments to be of commercial interest include F&B and commercial/industrial buildings, such as between interior and exterior spaces.

(10) FIGS. 1A-IF shows an exemplary conduit seal according to some of the embodiments. Collectively referring to FIGS. 1A-IF, the conduit seal **10** is shown with the individual components not assembled, including a through-hole connector **12** with a body **46** extending from a first end **14** to a second end **16**, the first end **14** having a connecting means **18** and the second end **16** having a flange **20**. The body **46** has an exterior surface, and the body **46** defines an internal cavity **48**. The second end **16** is longitudinally opposite the first end **14**. The conduit seal **10** further includes a first compliant member **22**, a second compliant member **24**, a locking member **26**, a first interface member **28**, and a second interface member **30**. FIGS. 1A-IF do not show a conducting member **36**, which can be seen in the cross-sectional view of the conduit seal **10** in FIG. 3. The first interface member **28**, the second interface member **30**, and the conducting member **36** are collectively referred to as an electrical conducting insert **38**.

(11) The first compliant member **22** and the second compliant member **24** are disposed circumferentially around the exterior surface of the through-hole connector **12**. In some embodiments, the conduit seal **10** includes the first compliant member **22** and the second compliant member **24**. In some embodiments, the conduit seal **10** includes either the first compliant member **22** or the second compliant member **24**, but not both the first compliant member **22** and the second compliant member **24**. The first compliant member **22** and the second compliant member **24** can be separate and distinct components. Alternatively, the first compliant member **22** and/or the second compliant member **24** can be made integral with another component of the conduit seal **10**. For example, the first compliant member **22** and the locking member **26** can be made as one integral component. Similarly, the second compliant member **24** and the flange **20** can be made as one integral component. When the second compliant member **24** and the flange **20** are not one integral component, the second compliant member **24** is connected to the flange **20**. The second compliant member **24** is configured to seal the flange **20** to a barrier (such as a barrier **40** shown in FIG. 3). There are advantages and disadvantages for having the first compliant member **22** and the second compliant member **24** as an integral or separate component of the conduit seal **10**. When the first compliant member **22** and the second compliant member **24** are integral with another component of the conduit seal **10**, some of the benefits include a compact form factor (ease of shipment and installation) for the conduit seal **10**. The compact form factor provides a distributor-friendly product. When the first compliant member **22** and the second compliant member **24** are not integral (i.e., separate and distinct) with another component of the conduit seal **10**, some of the benefits include the increased ability to use varied materials according to application requirements.

(12) The locking member **26** interfaces with the through-hole connector **12** via the connecting means **18**. The locking member **26** can be made of the same material as the through-hole connector **12**. The locking member **26** and the first compliant member **22**, whether or not they are integral, are configured to seal the locking member to a barrier (such as a barrier **40** shown in FIG. 3). The locking member **26** and the first compliant member **22** can be connected to one another and

configured to provide superior seal performance on a first side of a barrier (such as a first side **42** of the barrier **40** as shown in FIG. 3). The locking member **26** can be a variety of forms depending on the connecting means **18**. For example, if the connecting means **18** is a threaded interface, the locking member **26** can be a flange with a reciprocally threaded interface.

(13) The connecting means **18** can use a variety of means to connect the through-hole connector **12** to the locking member **26** including a threaded interface, mechanical fastener, clearance fit (such as bolt/shaft holes), interference fit (i.e., press fit or friction fit), transition fit (i.e., slip or push fit), or snap-fit (such as a spherical snap-fit, ball bearing snap-fit, slotted snap-fit, torsion snap joint, annular snap joint or cantilevered lug snap-fit). Set screw and compression are types of connecting means **18** that can be used for in a variety of applications, including electrical metal tubing (EMT). This list is not meant to be exhaustive and is only exemplary. The connecting means **18** can be threaded to allow for variation in a length that the conduit seal **10** needs to penetrate through a barrier (similar to a barrier **40** in FIG. 3) and for fixation to a structure, such as a barrier.

(14) FIG. 2 shows the conduit seal **10** according to some of the embodiments. FIG. 2 shows the conduit seal **10** of FIGS. 1A-IF fully assembled. Accordingly, similar reference numbers are used. Not shown in FIG. 2 is a barrier (similar to a barrier **40** in FIG. 3) between the first compliant member **22** and the second compliant member **24**. FIG. 2 shows a first end **32** of the conduit seal **10** and a second end **34** of the conduit seal **10**.

(15) FIG. 3 shows the conduit seal **10** according to some of the embodiments. FIG. 3 shows an embodiment for the conduit seal **10** with the electrical conducting insert **38**. Accordingly, similar reference numbers are used as FIGS. 1A-IF and FIG. 2. The electrical conducting insert **38** includes the first interface member **28**, the second interface member **30**, and the conducting member **36**. As shown in FIG. 3, the first interface member **28** is connected to the first end **14** of the through-hole connector **12**, the second interface member **30** is connected to the second end **16** of the through-hole connector **12**, and the conducting member **36** electrically connects the first interface member **28** to the second interface member **30**.

(16) The electrical conducting insert **38** provides electrical continuity through a barrier **40** from the first side **42** of the barrier **40** to a second side **44** of the barrier **40**. A first conduit can be connected to the first end **32** of the conduit seal **10**, and a second conduit can be connected to the second end **34** of the conduit seal **10**. The electrical conducting insert **38** enables electrical continuity between the first conduit connected to the first end **32** and the second conduit connected to the second end **34**. The first conduit and second conduit can be a variety of shapes, sizes, and materials depending on the application. For example, the first conduit and second conduit can be a cord as well as flexible or rigid.

(17) The barrier **40** can take a variety of forms. In some examples, the barrier **40** is a floor or a wall (e.g., a floor or a wall in a building). When the barrier **40** is a floor (e.g., a floor in a building), the first side **42** of the barrier **40** can be above or below the second side **44** of the barrier **40**. That is, the first side **42** of the barrier **40** can be located vertically above the second side **44** of the barrier **40**; alternatively, the second side **44** of the barrier **40** can be located vertically above the first side **42** of the barrier **40**. When the barrier **40** is a wall (e.g., a wall in a building), the first side **42** of the barrier **40** can be located horizontally across from the second side **44** of the barrier **40**. These examples of the barrier **40** are not exhaustive and meant to be exemplary only.

(18) Depending on conditions of the environment, it may be more beneficial to have the locking member **26** may be on the first side **42** or the second side **44**. For example, having the locking member **26** be located vertically below the flange **20** may make installation easier. That way, a user can extend the conduit seal **10** through the barrier **40** while keeping the flange **20** on the above floor side. The flange **20** will keep the conduit seal **10** from sliding through the hole of the barrier **40**. The user can then go down to the floor below and connect the locking member **26** to the connecting means **18** of the conduit seal.

(19) In contrast, if the conduit seal **10** was pushed through a hole in the barrier **40** while

maintaining the flange **20** on the below floor side, the conduit seal **10** may fall out of the hole because the flange **20** was not able to support the conduit seal **10**. This example is similar to the depiction shown in FIG. **3** where the barrier **40** is a floor/ceiling of a building.

(20) The barrier **40** can be any form or shape that divides the first side **42** of the barrier **40** from the second side **44** of the barrier **40**. For example, the barrier **40** can be an insulation panel. In addition, the first side **42** and the second side **44** can be positioned in any relation to one another: side by side; and above and below (and vice versa). Depending on the application, it may be beneficial to have the locking member **26** on the first side **42** or second side **44** of the barrier **40**.

(21) The electrical conducting insert **38** provides for electrical continuity between the first side **42** and the second side **44**. The electrical conducting insert **38** extends longitudinally within the internal cavity **48** along the through-hole connector **12** to electrically connect the first end **14** of the through-hole connector **12** to the second end **16** of the through-hole connector **12**. In some examples, the electrical conducting insert **38** can include more than one conducting member **36**. In some examples, the conducting member **36** does not extend substantially parallel to a longitudinal centerline of the through-hole connector **12**. Instead, the conducting member **36** can be helical or have a non-axial orientation, which may increase thermal resistance.

(22) In some examples, the first interface member **28** has a threaded portion and is configured to be connected to a separate conduit. Similarly, the second interface member **30** has a threaded portion and is configured to be connected to a second, separate conduit. In some examples, the interface members **28** and **30** are threadless instead of having threaded fittings.

(23) The conduit seal **10** can be made with a variety of materials according to application requirements, including: polymers, reinforced polymers, such as with additives, FDA approved materials and fire-resistant polymers. For example, fiber reinforced polymers can be used to achieve desired mechanical stiffness. FDA-approved polymers can be used for F&B applications. Intumescent material can be adopted if the conduit seal **10** application demands fire code compliance. In some examples, the intumescent material will expand and form carbonized porous foam when exposed to fire to seal the through hole and subsequently stop fire penetration from the first side **42** to the second side **44** of the barrier **40**. When the conduit seal **10** is a foamed polymer, the conduit seal **10** can have enhanced thermal resistance.

(24) In some examples, the conduit seal **10** can be made of a single material. Alternatively, some components of the device can be made of different materials. For example, the through-hole connector **12** and the interface members **28** and **30** can be made of a different material than one another. In some embodiments, the conduit seal **10** is injection molded via thermoplastic materials, compression (compression or transfer molding) molded via thermoset materials or potted. The present disclosure provides for the conduit seal **10** to include materials such as thermosetting plastics, silicone rubber gels, or epoxy resins. The conduit seal **10**, including the through-hole connector **12**, can be made with polymers through a variety of methods such as injection molding, compression molding, transfer molding, or potting.

(25) Examples of thermoplastic materials that may be used include one or more polymer composite materials, including but not limited to polyolefins, such as polypropylene, polyethylene, low density polyethylene, high density polyethylene, acetal and ketal based polymers and copolymers, polyesters (e.g. polyethylene terephthalate, polybutylene terephthalate), polycarbonate, polystyrene, polyether sulfone (PESU), polyphenylene sulfone (PPSU), polysulfone, and polytetrafluoroethylene (PTFE). Other polymers can also be implemented, including but not limited to polyvinyl chloride (PVC), polyetherimide (PEI), polyphenylene sulfide (PPS), polyether ether ketone (PEEK), polyphthalamide (PPA), polyoxymethylene (POM), phenolformaldehyde (PF), unsaturated polyester (UP), polyurethane (PUR and PU), polyimide, polyamide, polyvinyl alcohol, polyvinylidene chloride, polyacrylonitrile and polyalkylene paraoxybenzoate. Engineered polymeric materials can be used such as reinforced polymer materials and fire-resistant polymers including intumescent polymers.

(26) Examples of thermoset materials include but are not limited to epoxies, polyester, such as polyester bulk molding compounds (BMC), and phenolics. Rubbers, such as nitrile rubbers, butyl rubbers, neoprene, EPDM rubber (ethylene propylene diene monomer rubber), and silicone can also be used. The thermoset materials, including rubbers, can be used in a potted process. For example, the thermoset materials can be used in a potting process to encapsulate the interface members **28** and **30**.

(27) In some embodiments, the material of the conduit seal **10** can include a variety of types of fibers, including but not limited to carbon fiber, glass fiber, wool, silk, cotton, rayon, cellulose, cellulose acetate, flax, ramie, jute and aramid fibers. For example, the conduit seal **10** can be made of a polymeric composite with wool.

(28) In some embodiments, additives can be added to the materials of the conduit seal **10** to improve the qualities of the materials, including but not limited to the ultra-violet (UV) resistance, antimicrobial, conductivity, mechanical and thermal stability, chemical resistance, insulation property and decreased flammability. The additives can include, but are not limited to, ceramics, alumina, calcium silicate, flame retardants, expandable graphite and clays. Blowing agents can also be added to the formulation to achieve foaming structures. Possible blowing agents may be ammonium polyphosphate, melamine phosphate, urea, urea-formaldehyde resins, dicyandiamide, melamine or glycine. The additives may be in the range of 0% to 50%, or any intervening number or a smaller range within 0% to 50%, by mass in the conduit seal **10**. For example, the additives in the conduit seal **10** may be about 12% by mass or from about 15% to about 25%.

(29) The conduit seal **10** may need to be rigid or flexible depending on the shape of the barrier **40** and environmental conditions. In some examples, the conduit seal **10** can have a durometer range of about 40 Shore A to about 100 Shore A, or any intervening number. For example, the conduit seal **10** can have a durometer range of about 40 Shore A to about 90 Shore A.

(30) The conduit seal **10** may have a variety of shapes and sizes, depending on the barrier **40**, to achieve a proper seal. Although the cross-sectional shape of the conduit seal **10** is shown as cylindrical, other geometries (e.g., triangular or hexagonal) may be used. The geometry of the conduit seal **10** may be symmetrical or non-symmetrical. The exterior surface of the conduit seal **10** can be smooth or textured (e.g., corrugated). In some examples, such as in the F&B industry, it may be advantageous to have a smooth exterior. The shape and size of the conduit seal **10** can be selected based on the shape and size of the space of the barrier **40** that the conduit seal **10** is extending through. For example, if the length of the space of the barrier **40** is long, the conduit seal **10** will also need to be long. If the cross-sectional shape of the space of the barrier **40** is small, the conduit seal **10** will also need to be small. Components of the conduit seal **10** may have different sizes and/or geometries from one another. For example, the interface members **28** and **30** may have a circular cross-sectional shape, and the body **46** may have a square cross-sectional shape. The interface members **28** and **30** may have different cross-sectional shapes or sizes depending on the conduit that they are being connected to. As an example, the interface members **28** and **30** may both have a circular cross-sectional shape, but the first interface member **28** may be smaller than the second interface member **30** to connect to a smaller conduit.

(31) The conduit seal **10**, particularly the through-hole connector **12** and the locking member **26**, can have a low thermal conductivity. By having the conduit seal **10** be made of a material of low thermal conductivity, the heat leak or gain across a change in environment can be prevented or reduced. In turn, this may reduce HVAC demands for the environment and also prevent or reduce the issue of condensation, as previously described. The conduit seal **10**, including the through-hole connector **12**, can have a thermal conductivity ranging from about 0.2 Watt per meter-Kelvin to about 5 Watt per meter-Kelvin, or any intervening number. For example, the conduit seal **10** can have a thermal conductivity ranging from about 1 Watt per meter-Kelvin to about 3 Watt per meter-Kelvin, or about 0.5 Watt per meter-Kelvin.

(32) The first interface member **28** and the second interface member **30** (collectively referred to as

the interface members **28** and **30**) can be made of a variety of materials. In some examples, the interface members **28** and **30** are made of a conductive material or include a conductive material, such as a metallic material. Conductive materials include galvanized steel, stainless steel (304 or 316), aluminum, or red brass. In some examples, the interface members **28** and **30** may be made primarily of a non-conductive material and then include a conductive material so that the interface members **28** and **30** as a whole are sufficiently conductive.

(33) Similar to the interface members **28** and **30**, the conducting member **36** can be made of a variety of materials. The conducting member **36** can be made of or at least include a conductive material. In some examples, the conductive material includes copper, aluminum, or stainless steel. The conducting member **36** and the interface members **28** and **30** can be made of the same or substantially similar material. Alternatively, the conducting member **36** and the interface members **28** and **30** can be of a different material.

(34) The first compliant member **22** and the second compliant member **24** (collectively referred to as the compliant members **22** and **24**) can be made of a variety of materials. In some examples, the compliant members **22** and **24** are elastomeric, and elastomeric sealing is designed to provide superior and durable sealing to a structure. In some examples, the compliant members may not be needed when enhanced sealing to a structure is not required. Elastomeric materials include polyisoprene, polybutadiene (butadiene rubber), styrene-butadiene rubber, acrylonitrile-butadiene copolymer, isobutylene-isoprene copolymer, ethylene-propylene copolymer, polychloroprene, polysulfide, silicones (e.g., polydimethylsiloxane), polyurethane, polyacrylate elastomer, chlorinated polyethylene, styrene-isoprene-styrene copolymer (SIS), styrene-butadiene-styrene copolymer (SBS), ethylene-propylene-diene polymer, fluoroelastomers (e.g., tetrafluoroethylene propylene, fluorosilicone, and perfluoroelastomer), or any combination thereof.

(35) The conduit seal **10** can be assembled and manufactured through a variety of methods. The interface members **28** and **30** can be over-molded at each end of the through-hole connector **12**. The interface members **28** and **30** (e.g., over-molded metallic interfaces) can be a common conduit coupler or nipple, and electrically bonded with the conducting member **36** (e.g., a bonding conductor). The interface members **28** and **30** can have standard interface design (e.g., a National Pipe Taper (NPT) interface) so that the conduit seal **10** can be widely used on pipes and fittings.

(36) When installing the conduit seal **10**, the through-hole connector **12** can be inserted into the barrier **40** (e.g., the insulation wall, ceilings or insulation panels), and then the compliant members **22** and **24** (e.g., elastomer washers) are utilized on the flange **20** and the locking member **26** (e.g., a flange) to provide superior and durable sealing. Conduits can then be installed at the first end **14** and second end **16** of the through-hole connector **12**. In some examples, the locking member **26** can be a molded threaded locking cap to mechanically fix the conduit seal **10** to the barrier **40** (i.e., the structure the conduit seal **10** is intended to penetrate).

(37) The electrical conducting insert **38** can be over-molded via injection molding, compression molding, transfer molding, or potting. In manufacturing the conduit seal **10**, the electrical conducting insert **38** can be placed inside a mold, followed by molten polymer filling the cavity, encapsulating the electrical conducting insert **38** and forming the final geometry in one step. By changing the electrical conducting insert **38** (such as size, shapes and materials of the electrical conducting insert **38**), it is feasible to achieve multiple trade sizes in a single mold, reducing production cost. Manufacturing the conduit seal **10** can also incorporate scalable manufacturing (with limited capital expenditures) and automated production methods (e.g., for over-molding manufacturing). The electrical conducting insert **38** can also be added or installed after molding process via mechanical attachment or welding to the distinctive metallic components.

(38) FIG. 4 shows an exemplary flowchart according to some of the embodiments for securing a conduit seal to a barrier. The conduit seal can be any of the embodiments as described herein. The method **100** includes inserting **102** a first end of a through-hole connector through the barrier to a first side of the barrier. The method further includes maintaining **104** a flange connected to a

second end of the through-hole connector on a second side of the barrier. The method also includes sealing **106** the through-hole connector to the barrier by connecting a first compliant member connected to a locking member to the first side of the barrier and connecting a second compliant member connected to the flange to the second side of the barrier. In some examples, the locking member has a threaded interior surface and the first end of the through-hole connector has a threaded exterior surface, and the threaded interior surface of the locking member and the threaded exterior surface of the first end of the through-hole connector are reciprocally threaded. Connecting the first compliant member to the first side of the barrier can include screwing the threaded interior surface of the locking member onto the threaded exterior surface of the first end of the through-hole connector.

(39) The terminology used herein is intended to describe embodiments and is not intended to be limiting. The terms “a,” “an,” and “the” include the plural forms as well, unless clearly indicated otherwise. The terms “comprises” and/or “comprising,” when used in this Specification, specify the presence of the stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, and/or components.

(40) It is to be understood that changes may be made in detail, especially in matters of the construction materials employed and the shape, size, and arrangement of parts without departing from the scope of the present disclosure. This Specification and the embodiments described are examples, with the true scope and spirit of the disclosure being indicated by the claims that follow.

Claims

1. A conduit seal, comprising: a through-hole connector, wherein the through-hole connector includes: a body, wherein the body has an exterior surface, wherein the body defines an internal cavity, a first end, and a second end, wherein the second end is longitudinally opposite the first end; a compliant member, wherein the compliant member is disposed circumferentially around the exterior surface of the through-hole connector; a locking member, wherein the compliant member is connected to the locking member and is configured to seal the locking member to a barrier; and at least one electrical conducting insert comprising: a first interface member connected to the first end of the through-hole connector, the first interface member configured to connect to a first, separate conduit, a second interface member connected to the second end of the through-hole connector, the second interface member configured to connect to a second, separate conduit, and a conducting member electrically connecting the first interface member to the second interface member, wherein the at least one electrical conducting insert is located within the internal cavity defined by the body, and wherein the at least one electrical conducting insert extends longitudinally within the internal cavity along the through-hole connector to electrically connect the first end of the through-hole connector to the second end of the through-hole connector, wherein the first interface member has a first threaded portion and the second interface member has a second threaded portion.
2. The conduit seal of claim 1, further comprising a second compliant member, wherein the second compliant member is disposed circumferentially around the exterior surface of the through-hole connector.
3. The conduit seal of claim 2, wherein the second end of the through-hole connector comprises a flange, and wherein the second compliant member is connected to the flange and is configured to seal the flange to the barrier.
4. The conduit seal of claim 2, wherein the compliant member and the second compliant member are elastomeric.
5. The conduit seal of claim 2, wherein the compliant member and the second compliant member comprise polyisoprene, polybutadiene (butadiene rubber), styrene-butadiene rubber, acrylonitrile-

butadiene copolymer, isobutylene-isoprene copolymer, ethylene-propylene copolymer, polychloroprene, polysulfide, silicones, polyurethane, polyacrylate elastomer, chlorinated polyethylene, styrene-isoprene-styrene copolymer (SIS), styrene-butadiene-styrene copolymer (SBS), ethylene-propylene-diene polymer, fluoroelastomers, or any combination thereof.

6. The conduit seal of claim 1, wherein the first threaded portion of the first interface member is configured to be connected to the first, separate conduit, and wherein the second threaded portion of the second interface member is configured to be connected to the second, separate conduit.

7. The conduit seal of claim 1, wherein the first interface member and the second interface member comprise a metallic material.

8. The conduit seal of claim 1, wherein the conducting member comprises copper, aluminum, or stainless steel.

9. The conduit seal of claim 1, wherein the through-hole connector is injection molded, compression molded, transfer molded, or potted.

10. The conduit seal of claim 1, wherein the through-hole connector comprises a thermoplastic or a thermoset resin.

11. The conduit seal of claim 1, wherein the through-hole connector comprises an additive of ultra-violet (UV) resistance, antimicrobial, conductivity, mechanical and thermal stability, chemical resistance, insulation property or decreased flammability.

12. The conduit seal of claim 1, wherein a thermal conductivity of the through-hole connector ranges from about 0.2 Watt per meter-Kelvin to about 5 Watt per meter-Kelvin.

13. A method of securing a conduit seal to a barrier, comprising: inserting a first end of a body of a through-hole connector through the barrier to a first side of the barrier; maintaining a flange connected to a second end of the body of the through-hole connector on a second side of the barrier; sealing the through-hole connector to the barrier by connecting a first compliant member connected to a locking member to the first side of the barrier and connecting a second compliant member connected to the flange to the second side of the barrier, wherein the through-hole connector comprises at least one electrical conducting insert comprising: a first interface member connected to the first end of the through-hole connector, the first interface member configured to connect to a first, separate conduit, a second interface member connected to the second end of the through-hole connector, the second interface member configured to connect to a second, separate conduit, and a conducting member electrically connecting the first interface member to the second interface member, wherein the at least one electrical conducting insert is located within an internal cavity defined by the body, and wherein the at least one electrical conducting insert extends longitudinally within the internal cavity along the through-hole connector to electrically connect the first end of the through-hole connector to the second end of the through-hole connector, wherein the first interface member has a first threaded portion and the second interface member has a second threaded portion.

14. The method of securing the conduit seal to the barrier of claim 13, wherein the barrier is a floor in a building, and wherein the first side of the barrier is located vertically above the second side of the barrier.

15. The method of securing the conduit seal to the barrier of claim 13, wherein the barrier is a floor in a building, and wherein the second side of the barrier is located vertically above the first side of the barrier.

16. The method of securing the conduit seal to the barrier of claim 13, wherein the barrier is a wall in a building.

17. The method of securing the conduit seal to the barrier of claim 13, wherein the locking member has a threaded interior surface and the first end of the through-hole connector has a threaded exterior surface, wherein the threaded interior surface of the locking member and the threaded exterior surface of the first end of the through-hole connector are reciprocally threaded, and wherein connecting the first compliant member to the first side of the barrier comprises screwing

the threaded interior surface of the locking member onto the threaded exterior surface of the first end of the through-hole connector.

18. The method of securing the conduit seal to the barrier of claim 13, further comprising connecting a first conduit to a first end of the conduit seal.

19. The method of securing the conduit seal to the barrier of claim 13, further comprising connecting a second conduit to a second end of the conduit seal.
